

Multiple Choice (10 marks)

1. The lithosphere of a planet is the layer that consists of
 - (a) the softer rocky material of the mantle.
 - (b) the lava that comes out of volcanoes.
 - (c) material between the crust and the mantle.
 - (d) the rigid rocky material of the crust and uppermost portion of the mantle.
2. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
3. If the Moon is setting at noon the phase of the Moon must be
 - (a) third quarter.
 - (b) waning crescent.
 - (c) waxing crescent.
 - (d) full.
4. You are pushing a truck along a road. Would it be easier to accelerate this truck on Mars? Why? (Assume there is no friction)
 - (a) It would be harder since the truck is heavier on Mars.
 - (b) It would be easier since the truck is lighter on Mars.
 - (c) It would be harder since the truck is lighter on Mars.
 - (d) It would be the same no matter where you are.
5. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
 - (a) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) metals rocks
 - (b) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) rocks metals
 - (c) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) metals rocks
 - (d) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) rocks metals

True / False (10 marks)

Answer True or False

6. True or False: Planetary rings are primarily composed of dust grains that escape from passing comets and accumulate around planets.
7. True or False: Venus exhibits evidence of surface processes including impacts, erosion, and volcanism.
8. True or False: Sirius is the second closest star system to Earth, after Proxima Centauri/Alpha Centauri.
9. True or False: The Sun's corona is too diffuse to be seen clearly, except during a total solar eclipse when the bright photosphere is obscured.
10. True or False: In 20000 years, the Moon will be further from the Earth and the Earth's rotation period will increase, resulting in longer days.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of Betelgeuse as a prominent star in the night sky, including its characteristics and its role within the Orion constellation.
12. Discuss the implications of Kepler's laws of planetary motion and analyze how they relate to gravitational attraction, particularly addressing concepts not directly derived from these laws.

End of Paper